

Scientific Communities in the Literature Sample

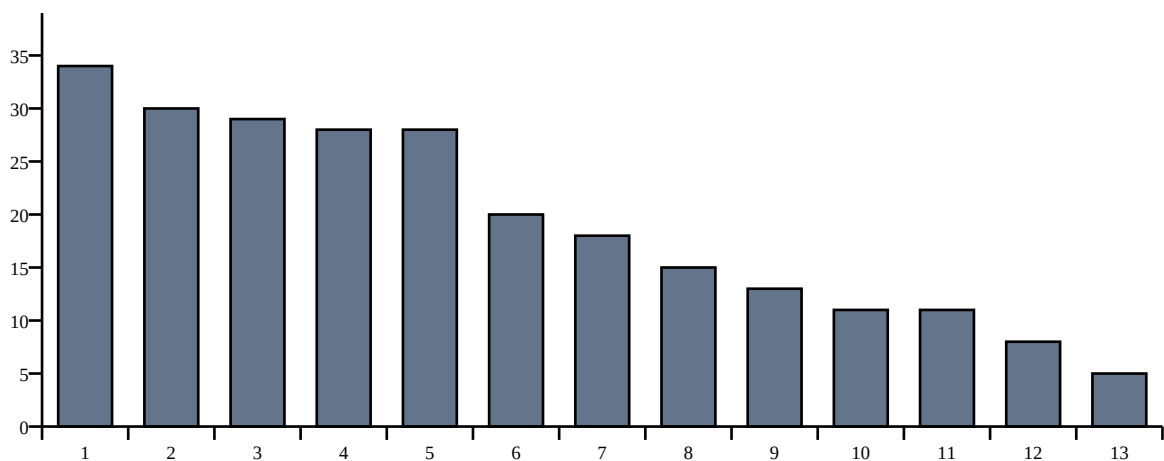
Inductive qualitative categorization based on the CSV fields **title** and **keywords**.

Dataset: **250** records from *data_with_links.csv*. The coding focuses on identifying thematic scientific communities represented in the sample. Counts are indicative primary assignments; many articles could also be coded as secondary cases in adjacent communities.

Coding Procedure

I combined the title and keyword fields, inspected recurring terms and themes, and formed a compact set of inductive community codes. Each record was assigned to the most strongly represented category based on title/keyword evidence. The resulting codebook is intended for transparent overview coding, not as a substitute for full-text manual coding.

Primary category counts (numeric labels correspond to the overview table)



Overview of Generated Categories

#	Community category	n
1	Teacher Education, Professional Development, and Research Use	34
2	Assessment, Psychometrics, Testing, and Academic Integrity	30
3	Higher Education, Doctoral Education, and Academic Careers	29
4	Learning Analytics, AI, and Data-Driven Educational Technology	28
5	STEM, Mathematics, Science, and Engineering Education	28
6	Reading, Literacy, Writing, and Language Education	20
7	Motivation, Well-Being, Self-Regulation, and Educational Psychology	18
8	Other / Cross-cutting Education Research	15
9	Inclusion, Equity, Diversity, and Special Education	13
10	Open Science, Meta-Research, Bibliometrics, and Evidence Synthesis	11
11	Curriculum, Instructional Design, and Pedagogical Interventions	11
12	COVID-19, Remote Learning, and Educational Disruption	8
13	Immersive, Virtual, Augmented, and Simulation-Based Learning	5

Category Definitions and Examples

1. Learning Analytics, AI, and Data-Driven Educational Technology (n = 28)

Definition. Research communities using artificial intelligence, machine learning, learning analytics, data mining, generative AI, or computational models to analyze, support, or automate educational processes.

Example title. On the Use of eXplainable Artificial Intelligence to Evaluate School Dropout

Example keywords. xplainable artificial intelligence; XAI; school dropout; machine learning; educational data science

2. STEM, Mathematics, Science, and Engineering Education (n = 28)

Definition. Communities centered on STEM learning and teaching, including mathematics, science, biology, engineering, technology, modeling, CAD, science communication, and STEM interventions.

Example title. Gamifying Renewable Energy: Enhancing Pre-University Students Knowledge and Attitudes Toward Science and Technology

Example keywords. escape room; science and technology (S&T); science, technology, engineering, and mathematics (STEM); sustainability education

3. Reading, Literacy, Writing, and Language Education (n = 20)

Definition. Scholarship focused on reading fluency, comprehension, writing, vocabulary, language learning, literacy assessment, and text-based learning processes.

Example title. Development and validation of a rapid and precise online sentence reading efficiency assessment

Example keywords. dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics

4. Teacher Education, Professional Development, and Research Use (n = 34)

Definition. Communities studying teachers, pre-service teacher preparation, teacher competencies, professional learning, evidence-informed practice, and school leadership.

Example title. Rethinking Professional Development: Development and Evaluation of an Evidence-Based Online PD Course on the Effective Use of Technology in the Classroom

Example keywords. online professional development; ICAP; in-service teacher training; prepost design; teachers technological pedagogical knowledge

5. Higher Education, Doctoral Education, and Academic Careers (n = 29)

Definition. Research on universities, college students, undergraduate and graduate education, doctoral training, campus experiences, career outcomes, and postsecondary transitions.

Example title. Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education

Example keywords. graduate education; doctoral training; program evaluation; career outcomes; taxonomy; PhD

6. Inclusion, Equity, Diversity, and Special Education (n = 13)

Definition. Communities concerned with inclusive education, special education, disability, equity, gender, underrepresented groups, access, diversity, rurality, and social justice in education.

Example title. Effect of Local Norms on Racial and Ethnic Representation in Gifted Education

Example keywords. assessment; descriptive analysis; diversity; equity; gifted education; hierarchical linear modeling; policy analysis

7. Assessment, Psychometrics, Testing, and Academic Integrity (n = 30)

Definition. Research communities developing, validating, or evaluating assessments, psychometric instruments, tests, measurement models, academic integrity systems, and fairness of evaluation.

Example title. A New Measurement Instrument for Music-Related Argumentative Competence: The MARKO Competency Test and Competency Model

Example keywords. music-related argumentation; competency; assessment; music; reasoning; item response theory; empirical research; musical judgment

8. COVID-19, Remote Learning, and Educational Disruption (n = 8)

Definition. Work examining pandemic-related disruption, lockdown, emergency remote teaching, home education, stress, technology-mediated schooling, and COVID-era educational change.

Example title. Childrens Motivation to Learn at Home During the COVID-19 Pandemic: Insights From Indian Parents

Example keywords. motivation; COVID-19; learning; survey; India; parent; children

9. Motivation, Well-Being, Self-Regulation, and Educational Psychology (n = 18)

Definition. Educational psychology communities focusing on motivation, self-efficacy, engagement, emotions, wellbeing, mental health, personality, self-regulation, and learner identity.

Example title. Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources

Example keywords. COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories

10. Open Science, Meta-Research, Bibliometrics, and Evidence Synthesis (n = 11)

Definition. Meta-scientific communities studying open science practices, data sharing, preregistration, bibliometrics, systematic reviews, meta-analyses, research methods, and evidence mapping.

Example title. Open Education Science

Example keywords. preregistration; open science; registered report; open access

11. Immersive, Virtual, Augmented, and Simulation-Based Learning (n = 5)

Definition. Communities researching virtual reality, augmented reality, metaverse environments, simulations, immersive technologies, serious games, and technology-enhanced experiential learning.

Example title. Integrating immersive technologies with STEM education: a systematic review

Example keywords. immersive technologies; virtual reality; augmented reality; STEM education; PICOS approach; PRISMA guidelines; teaching and learning

12. Curriculum, Instructional Design, and Pedagogical Interventions (n = 11)

Definition. Studies of curriculum design, instructional models, learning designs, classroom interventions, pedagogy, course redesign, and experimentally evaluated teaching approaches.

Example title. Empowering Future HR Professionals: A Design-Based Research Approach to Project-Based Learning in Work and Organizational Psychology

Example keywords. project-based learning; open pedagogy; higher education development; human resource development; design-based research; student-centered learning; psychology education; openness; academic culture transformation

13. Other / Cross-cutting Education Research (n = 15)

Definition. Residual or cross-cutting education studies whose titles and keywords do not clearly map to one of the larger communities above. These cases should be reviewed manually if a fully exhaustive codebook is required.

Example title. Tone superimposition technique in Speech Sciences: a tutorial

Example keywords. tone superimposition; speech sciences; Mandarin lexical tones; stimuli; tutorial

Interpretive Summary

The sample is dominated by several overlapping education-research communities. The largest visible clusters are teacher education and professional development, assessment and psychometrics, higher education, STEM education, and AI/learning analytics. Smaller but analytically important communities include open science/meta-research, immersive learning technologies, COVID-era educational disruption, and inclusion/equity research.

Because the coding uses only title and keyword metadata, some communities may be undercounted when relevant concepts are present only in abstracts or full texts. Conversely, broad terms such as assessment, evaluation, design, and technology can connect several communities. For a final publication-quality coding scheme, I recommend a second manual review of borderline and multi-topic articles.

Traceability Note

A companion CSV with the primary category assignment, match score, and matched keywords was generated for auditability: `community_category_assignments.csv`.